

AAI-500 – Team Project Status Update Form

Fill out this form and submit it by the end of Module 4 in Blackboard.

Team Number: **Group 4**

Team Leader/Representative: **Ian Schmitt**

Full Names of Team Members:

1. **Keana Gindlesperger**
2. **Ian Schmitt**
3. **Guna Pasupathy**

Title of Your Project: **Predicting 30-Day Heart Failure Readmission Risk Using Clinical, Treatment, and Social Determinant Data**

Short Description of Your Project and Objectives: **This project will analyze factors associated with 30-day heart failure readmission and build a defensible risk-prediction workflow. Objectives: clean and validate the data, perform EDA and statistical testing, compare interpretable and machine-learning classification models against a baseline, and translate results into practical recommendations while clearly documenting limitations.**

Name of Your Selected Dataset: **Heart Failure Readmission and Social Determinants of Health Dataset**

Description of Your Selected Dataset (data source, number of variables, size of dataset, etc.):
The selected dataset is from Kaggle and contains 3,000 patient records with 16 variables. It includes clinical factors such as age, BMI, BNP, sodium, creatinine, blood pressure, and heart rate; treatment factors such as ACE inhibitor, beta blocker, and diuretic use; and social factors such as income level and distance to the hospital. The target variable is readmitted 30d, indicating 30-day readmission status.

Initial EDA found a reasonably balanced target (~41% readmitted), no duplicate rows, low missingness (3% in BMI, sodium, and creatinine), and patient id as an identifier to exclude from modeling. Data-quality checks flagged impossible negative creatinine values and several extreme clinical values that require review before modeling.

Are you using and practicing GitHub as a code hosting platform for version control and collaboration? If yes, provide the link here: <https://github.com/KeanaGindy/AAI-500-Final-Project>

How many times have your members met in the last two weeks? **Twice, with additional ongoing communication through Slack.**

What was the agreed upon method of communication? Are you using any teamwork project management software, such as [Trello](#) or [Asana](#)? **No teamwork project management software is being used. The group is actively communicating on slack to discuss project responsibilities.**

Comments/ Roadblocks: **Group plans to coordinate to discuss scheduling, distribution of responsibilities, project structure, and workflow in greater detail this week.**